

MBL 331 1.0

EXTREMOPHILES

**Isolation and Identification of Halophilic Archaea and Bacteria
for Screening and Characterization of Extremozymes**

Sampling Zone – Intermediate Zone



Date - 2026.09.14

Experiment name - Isolation and Identification of Halophilic Archaea and Bacteria for Screening and Characterization of Extremozymes

Objectives

1. Field Objectives:

- ✓ To observe and document salt production stages at Hambantota Salt Factory
- ✓ To collect water, sediment, and salt crystal samples from varying salinity zones of the salt pan. .

2. Laboratory Objectives:

- ✓ To isolate and culture halophiles from collected salt pan samples.
- ✓ To record morphological and growth characteristics of all isolates.
- ✓ To conduct amylase and protease activity assays on halophilic isolates

Abstract

Halophiles are extremophilic microorganisms that require higher sodium chloride concentrations for growth and survival, and they represent a promising source of salt-tolerant enzymes (extremozymes) with industrial potential. This study aimed to isolate and characterize halophilic microbes from the intermediate ponds (Zone 2) of the Hambantota Salt Pan and to screen the isolates for extracellular amylase and protease activity under high-salinity conditions. Water, sediment, and salt crystal samples were collected aseptically, and physicochemical parameters were recorded on-site (temperature 33 °C, salinity 105 ppt, pH 8.1, conductivity 180 mS, internal density 14 g L⁻¹). Halophiles were isolated by serial dilution and spread-plating onto Marine agar supplemented with 10% and 25% (w/v) NaCl, followed by incubation at 37–42 °C for 3–4 days. Three distinct isolates were purified and characterized by colony morphology and Gram staining (Isolate 1 (Gram-positive rod), Isolate 2 (Gram-negative coccus), and Isolate 3 (curved rod with light orangish-pink appearance). Amylase activity was assessed on starch agar flooded with iodine solution, while protease activity was assessed on skim milk agar with TCA (trichloroacetic acid). Isolates 1 and 2 were positive for amylase, whereas Isolate 3 was negative. All three isolates demonstrated protease activity. These results indicate that the intermediate ponds of the Hambantota Salt Pan harbour metabolically diverse halophiles capable of producing salt-stable extracellular enzymes with potential applications in various industries like food processing, environmental management and biotechnology. The study highlights the biotechnological value of Sri Lankan hypersaline environments and supports future molecular identification and enzyme optimization studies.

Keywords: halophiles, hypersaline environments, extremozymes, amylase, protease

Introduction

Extremophiles are the organisms that have been evolutionary adapted to survive and replicate under harsh environmental conditions normally lethal to other forms of life including extreme temperatures, pH levels, pressure, radiation or high salt concentrations. Among these halophiles represent a distinct group of extremophiles that require high concentrations of sodium chloride for cellular stability, metabolic enzyme functionality, and growth. Halophiles can be classified based on their optimal growth salinity level as slight halophiles (1% – 3% NaCl (w/v)), moderate halophiles (3% – 15% NaCl (w/v)) and extreme halophiles (15% – 30% NaCl (w/v)). There are specifically two groups of halophiles as halophilic bacteria and halophilic archaea. Their ability to survive in high salinity stress is heavily shaped by their divergent osmoregulatory strategies. Halophilic bacteria predominantly employ the “compatible solute” strategy. They synthesize or employ small, organic uncharged molecules like glycine betaine, ectoine and trehalose. These solutes accumulate in the cytoplasm of these organisms to match the external osmotic pressure and disrupting the conformational foldings and electrostatic interactions of intracellular proteins. However this strategy is energetically expensive. It diverts a significant amount of ATP and metabolic precursors away from growth. So this often limits their competitive dominance at highest salinities. Conversely halophilic archaea utilize the less energy consuming “salt-in strategy”. In this method massive quantities of K^+ ions are pumped into the cytoplasm to achieve osmotic balance. However their proteins must have special structural adaptations to efficiently function in near saturating KCl conditions. They exhibit a distinct amino acid composition with a high frequency of negatively charged amino acids (aspartate, glutamate) on their solvent exposed surfaces. These negative charges attract water molecules forming a hydration shell around the proteins and prevent protein aggregation and clumping (salting-out). This difference in protein chemistry suggests that archaeal extremozymes are uniquely stable and active at non-aqueous environments or low-water activity reactions.

Located in the semi-arid zone of southern Sri Lanka, the Hambantota Salt Pan managed by Lanka Salt Limited is an ecologically important human-made hypersaline system. Driven by high ambient temperatures, high solar radiation and persistent strong winds, sea brine undergoes progressive evaporation, causing sequential precipitation of minerals according to their solubility limits. Beyond its economic role as a major national producer of sodium chloride, the Hambantota saltern serves as a unique dynamic model for studying haloextremophilic microbiology, hypersaline zonation, and poly-extremophilic enzyme activity. The saltern operates through a controlled hydrodynamic flow where seawater is sequentially pumped or gravity-fed through a chain of shallow earthen ponds. As volume diminishes through solar evaporation, the concentration of total dissolved solids rises sharply from seawater (~3.5% NaCl) to saturated brine (>30% NaCl). Based on the salinity gradient and physical characteristics of the saltern five major ecological sampling sites were identified for this study. In Zone 1 (Inlet Water- low salinity), freshly pumped coastal seawater begins to evaporate and causes calcium carbonate to precipitate. This zone supports typical marine communities like diatoms, eukaryotic phytoplankton, and moderate marine bacteria. As water volume drops in Zone 2 (Intermediate Ponds- medium salinity), gypsum precipitates to form benthic crusts. Here, typical marine life gives way to moderately halophilic bacteria (*Halomonas*), halotolerant fungi, and the microalga *Dunaliella salina*, which produces high levels of beta-carotene. In Zone 3 (Crystallizer Ponds- high salinity), saturated brine leads to pure NaCl precipitation under very low level of oxygen, where extreme haloarchaea (*Halobacterium*, *Haloferax*) and specialized bacteria (*Salinibacter ruber*) dominate, turning the water red with bacterioruberin pigments that enhance thermal solar absorption. Post-harvest Zone 4 (Salt Heaps & Factory evaporation Beds) represents a solid-phase environment with extremely low water activity. It harbors stress-resilient haloarchaea, halophilic bacteria and microbes associated with surface-bound biofilms. Finally, Zone 5 (Rhizosphere Soil) forms a dynamic terrestrial-aquatic

interface around salt marsh halophytes (*Salicornia*, *Suaeda*). Here fluctuating salinities support Plant Growth-Promoting Rhizobacteria (PGPR) under salinity stress.

In the field first the spatial coordinates of the sampling station and then the basic water and sediment data like temperature, pH and salinity were recorded. Samples were collected under aseptic conditions and were transported to the laboratory. Once in the laboratory differential media and osmotic conditions were applied to isolate the halophiles. Colony morphology, growth rates, and pigmentations of the grown isolates were recorded afterwards. Finally pure isolates were screened for industrially important extracellular enzymes (amylase and protease) under high salinity conditions.

This overall study provides important insights into extremophilic survival mechanisms in hypersaline environments and it combines fundamental microbial physiology with practical biotechnological applications.

MATERIALS

- Sterile 250 ml. sampling bottles
- Sterile polyethylene/zip-lock bags
- Sterile spatulas and scoops
- GPS device/smartphone with GPS
- thermometer
- pH meter
- Cooler box with ice packs
- Permanent markers and waterproof labels
- Disposable gloves
- Hand sanitizer

Methodology

Hambantota Salt Pan & Salt Factory

Sampling Zone :

ZONE 2 – Intermediate Ponds (Medium Salinity)

Procedure :

The field sampling was carried out at the intermediate ponds (Zone 2) of the Hambantota Salt Pan, which represented a medium-salinity environment. The GPS coordinates of the sampling site were recorded. The temperature of the sampling site was measured using a thermometer.

A 250 mL water sample was collected into a sterile sampling bottle under aseptic conditions. Approximately 100 g of sediment was collected using sterile equipment and transferred into a sterile polyethylene bag. Approximately 100 g of salt crystals were also collected into a sterile container. All collected samples were labelled with the date, group number, site code, and sample type. The samples were then stored in a cool box and transported to the laboratory for further analysis.

Isolation of Halophiles

Isolation of Halophilic Bacteria

First, Temperature, opacity, pH and salinity of the water and sediment samples were measured. Then, collected samples were vortexed. Marine agar plates containing 10% and 25% NaCl were provided. Serial dilution series of the collected water and sediment sample were prepared , and the 10^{-2} , 10^{-3} , and 10^{-4} dilutions were plated separately onto both 10% and 25% NaCl Marine agar plates. A 0.1 mL aliquot of each dilution was spread evenly over the surface of the respective agar plates. The inoculated plates were incubated at 37 – 42 °C for 3 – 4 days.

After incubation, the developed colonies were observed based on colony morphology, pigmentation, and growth. Three distinct isolates showing growth on the culture plates were selected. Fresh Marine agar plates were prepared, and the selected three isolates were streaked onto the plates to obtain isolated colonies. The streaked plates were incubated under the appropriate conditions.

Amylase Activity Assay

Starch agar medium containing 10–20% NaCl was prepared for the detection of amylase activity. The three selected isolates were inoculated onto the starch agar plates and incubated for 3–5 days. After incubation, the plates were flooded with iodine solution, and the formation of clear zones around the colonies was observed. The clear zones were considered indicative of starch hydrolysis and amylase activity.

Protease Activity Assay

Skim milk agar medium containing 10–20% NaCl was prepared for the detection of protease activity. The three selected isolates were inoculated onto the skim milk agar plates and incubated for 3–5 days. After incubation, TCA (trichloroacetic acid) was added to the plates to observe the zones produced around the colonies. The formation of clear zones was observed and recorded as an indication of proteolytic activity.

Observations

Sampling site



Figure 1 – Salt evaporation ponds at the Hambantota salt pan



Figure 2 – Pyramid shaped salt heaps stored at the Hambantota salt pan



Figure 1 - Pink colour crystallizer pond observed at the Hambantota salt pan



Figure 4 – Collection of the samples and measuring parameters from an intermediate pond at the Hambantota salt pan

Bacterial Isolates

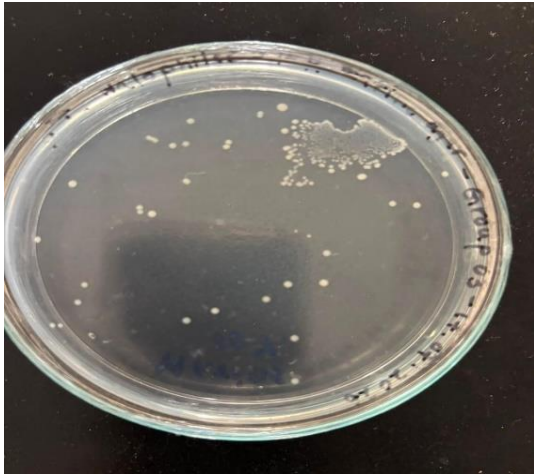


Figure 5 - Halophilic bacterial growth on 25% NaCl Marine agar at 10^{-3} dilution

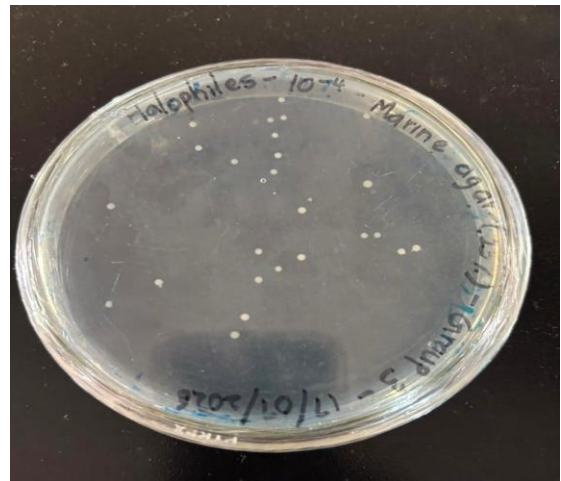


Figure 6 - Halophilic bacterial growth on 10% NaCl Marine agar at 10^{-4} dilution

Microscopic view of bacterial isolates (Gram-Staining)

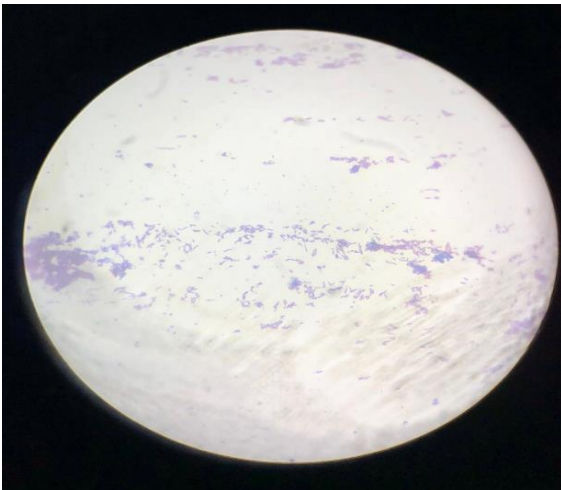


Figure 7 – Gram-staining of bacterial isolate 1

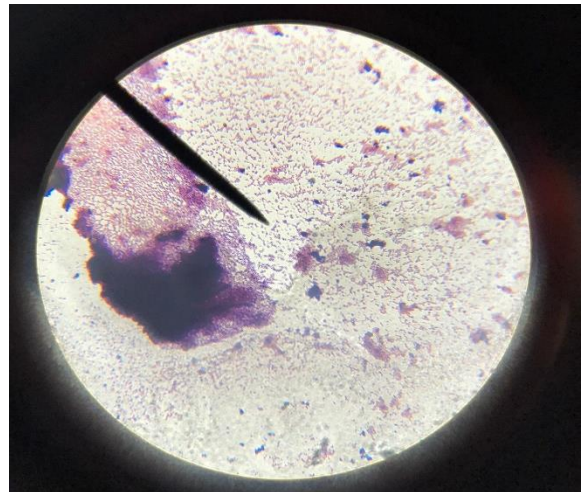


Figure 8 – Gram-staining of bacterial isolate 2

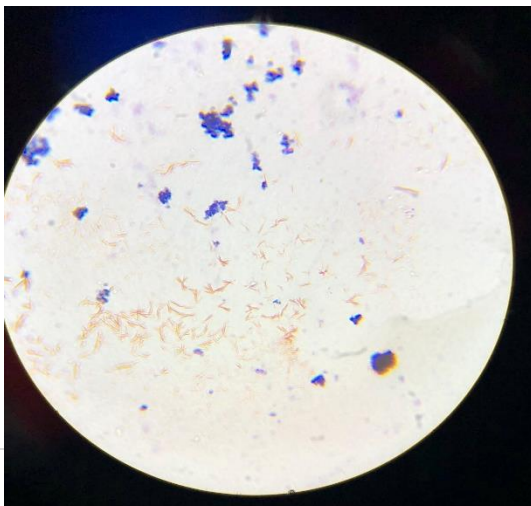


Figure 9 – Gram-staining of bacterial isolate 3

Amylase Activity

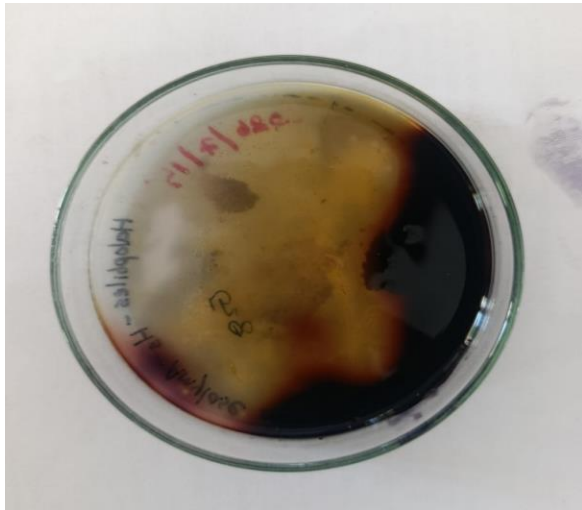


Figure 10 – Amylase activity of the bacterial isolate 1



Figure 11 - Amylase activity of the bacterial isolate 2



Figure 12 - Amylase activity of the bacterial isolate 3

Protease Activity



Figure 13 – Protease activity of bacterial isolate 1

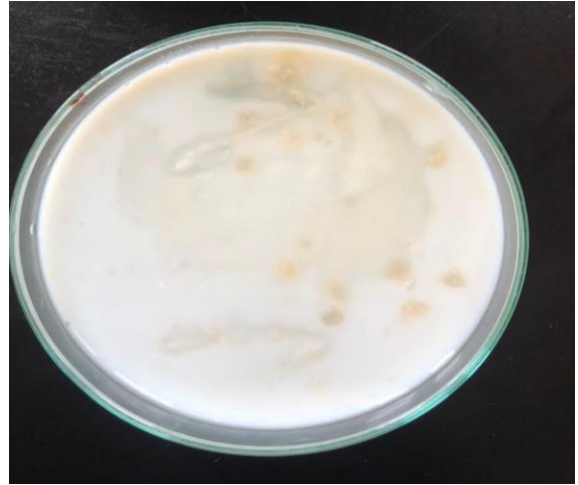


Figure 14 - Protease activity of bacterial isolate 2



Figure 15 - Protease activity of bacterial isolate 3

Results

Hambantota Salt Pan & Salt Factory

Sampling Zone :

ZONE 2 – Intermediate Ponds (Medium Salinity)

Temperature – 33 °C

Internal Density – 14 gL⁻¹

Salinity – 105 ppt

pH – 8.1

Conductivity – 180mS

Colony Morphology

Isolate	Colony colour	shape	Margin	Texature
Isolate 1	White	Circular	Entire	Smooth, shiny
Isolate 2	White	Circular	Entire	Smooth, opaque
Isolate 3	yellow	Circular	wavy	Smooth, pigmented

Microscopic view of bacterial isolates (Gram - Staining)

Isolate	Gram positive or negative	Shape
Isolate 1	Positive	rod
Isolate 2	Negative	coccus
Isolate 3	Appeared as a light orangish pink	curved rod

Amylase Activity

Bacterial Isolates	Amylase Activity
Isolate 1	Positive
Isolate 2	Positive
Isolate 3	Negative

Protease activity

Bacterial Isolates	Amylase Activity
Isolate 1	Positive
Isolate 2	Positive
Isolate 3	Positive

Discussion

Environmental Context and Sampling Site Dynamics

The microbial samples analyzed in this study were retrieved from the intermediate tanks of the Hambantota salt pan, an engineered solar saltern functioning as a pronounced hypersaline ecosystem. Environments of this nature subject indigenous microbial communities to severe physiological stress factors, including extreme osmotic pressure, high ionic concentrations (particularly sodium and chloride ions), elevated ambient temperatures, and intense solar radiation.

These harsh abiotic conditions categorize the salt pan as an extremophilic habitat, which naturally restricts biodiversity while enriching for specialized halophilic or halotolerant microorganisms. To survive, these organisms have evolved unique structural and biochemical adaptations, such as compatible solute accumulation, specialized cell wall compositions, and salt tolerant enzymatic machinery, making them exceptional candidates for ecological and biotechnological exploration.

Sampling Methods and Aseptic Controls

Field collection procedures were meticulously designed to preserve native microbial integrity while preventing environmental contamination.

- **Aseptic Precautions:** The utilization of presterilized sampling bottles, zip-lock bags, and sanitized scoops ensured that only the resident microflora of the intermediate tank were captured, effectively blocking external contaminants from instruments or handlers.
- **Environmental Tracking:** Parameters such as temperature, pH, and salinity were recorded on-site to document the exact physicochemical baseline governing the niche.
- **Transport:** Storing the collected water, sediment, and salt crystal samples in a styrofoam box minimized metabolic shifts or thermal shock during transit back to the laboratory, keeping the microbial populations in a stable state prior to processing.

Culturing Methodology and Isolation Challenges

To isolate cultivable halophiles, the samples underwent vigorous vortexing and serial dilution, followed by spread-plating onto Marine Agar enriched with high concentrations of 10% and 25% NaCl. These elevated salt concentrations served as a selective pressure, suppressing non-halophilic mesophiles and encouraging the proliferation of halophilic strains.

- **Primary Plate Observations:** Initial spread plates displayed a high density of diverse microbial colonies. Several smaller, pigmented colonies were visible early on; however, by the time isolation was attempted, these pigmented variants had dried out, and their respective isolated plate showed signs of contamination.
- **Successful Isolation:** Due to these contamination setbacks and desiccation issues, only three distinct, uncontaminated colonies were successfully rescued and sub-cultured for enzymatic activity evaluation.

Morphological and Microscopic Characterization of Isolates

Microscopic examination and Gram-staining of the three purified isolates revealed distinct cellular morphologies, highlighting micro-diversity even within a small subset of survivors:

- **Isolate 1:** Characterized as a Gram positive, rod-shaped bacterium.
- **Isolate 2:** Identified as a Gram negative coccus.
- **Isolate 3:** Appeared as a light orangish pink curved rod. Interestingly, despite its subtle hue, the definitive Gram reaction status of this isolate could not be reliably resolved under standard staining procedures. Furthermore, macroscopic inspection of the final isolated cultures showed a lack of pronounced pigmentation compared to the fleeting colored colonies observed on the original primary plates.

Enzyme Assays: Amylase and Protease Profiles

To evaluate the metabolic versatility of the isolates, extracellular enzymatic screening was performed under elevated salt conditions. Saline-adapted enzymes are of significant industrial interest because standard enzymes typically lose structural stability and catalytic activity in high ionic strength environments.

1. Amylase Activity (Starch Hydrolysis)

- **Principle:** Starch agar plates inoculated with the isolates were flooded with iodine solution following incubation. Starch reacts with iodine to form a dark blue black complex, whereas areas where extracellular amylase has hydrolyzed starch remain clear.
- **Results:** Isolate 1 and Isolate 2 both tested positive, evidenced by distinct clear hydrolysis halos surrounding their growth zones. Conversely, Isolate 3 tested negative, showing no surrounding halo against the dark iodine background.

The presence of amylase in the majority of the isolates (Isolates 1 and 2) indicates that carbohydrate-degrading pathways are actively utilized within this niche, likely for scavenging complex organic carbon sources derived from dead organic matter or primary producers in the saltern. The contrast provided by Isolate 3 which lacked amylase activity demonstrates clear metabolic partitioning within the microbial community. This suggests that

Isolate 3 relies on alternative carbon and energy acquisition strategies, avoiding direct competition for starch substrates

2. Protease Activity

- **Principle:** Skim milk agar contains casein, which gives the medium an opaque white appearance. Protease secretion breaks down casein into soluble peptides, visualized by adding trichloroacetic acid (TCA) or observing clear zones directly.
- **Results:** All three isolates demonstrated positive proteolytic activity, successfully forming clear zones of degradation on the milk agar.

presence of proteolytic capability across all tested strains points to a high ecological demand for proteinaceous nitrogen sources within the intermediate tank environment. Because free amino acids and peptides are vital for osmoadaptation serving as precursors for compatible solutes, the ability to break down complex proteins is a crucial survival mechanism. From a biotechnology perspective, discovering robust, salt-tolerant proteases across multiple distinct morphological types from a single sampling site highlights the intermediate tank as a promising reservoir for industrial enzymes. These halophilic proteases possess strong potential for applications requiring high ionic stability, such as in industrial detergent formulations, leather processing, and marine biotransformations.

Comparative Analysis and Biotechnological Implications

Comparing the functional profiles of the three isolates demonstrates a notable metabolic capability within this hypersaline community. While carbohydrate degrading amylase activity was found only in Isolates 1 and 2, protein degradation via protease production was shared by all three strains. The successful expression of extracellular amylases and proteases under high-salt conditions holds substantial promise for industrial biotechnology. Conventional enzymes derived from mesophilic organisms typically denature, aggregate, or lose catalytic efficiency in the presence of high ionic strengths or extreme osmotic pressures. In contrast, halozymes are structurally adapted often featuring an abundance of acidic amino acids on their surfaces to maintain structural integrity and catalytic function in harsh environments. Detergent Formulations, Halotolerant proteases are highly sought after as additives in commercial laundry and automatic dishwashing detergents, where they must remain stable in high ionic strength formulations containing builders and surfactants. Textile and Starch Processing: Halotolerant amylases offer distinct advantages in textile desizing and starch liquefaction processes, particularly where saline wastewater recycling or high-temperature, high-salt processing conditions are employed. Food and Leather Industry: Proteases functioning efficiently under saline conditions can be

utilized in leather processing (bating hides in high-salt curing brines) and food fermentations, reducing the need for extensive desalting steps and minimizing freshwater consumption.

Future research should focus on molecular identification via 16S rRNA gene sequencing and quantitative optimization of these enzymes to fully realize their commercial viability.

Conclusion

In this study, halophilic bacteria were successfully isolated from the intermediate pond (Zone 2) of the Hambantota Salt Pan under high-salinity conditions. The sampling site showed a temperature of 33 °C, salinity of 105 ppt, pH of 8.1, and conductivity of 180 mS. Three distinct bacterial isolates were successfully purified and characterized based on their colony and microscopic morphology. The isolates showed different morphological characteristics, indicating microbial diversity within the intermediate-salinity environment.

The enzymatic screening showed that Isolates 1 and 2 exhibited amylase activity, while all three isolates showed protease activity. These results demonstrate that the isolated microorganisms were capable of producing extracellular enzymes under high-salt conditions. The presence of salt-tolerant protease activity in all three isolates and amylase activity in two isolates indicates their potential importance in biotechnological applications where enzymes are required to function under saline or stressful conditions.

Although some difficulties such as contamination and desiccation of early pigmented colonies were encountered during isolation, three uncontaminated isolates were successfully recovered for further analysis. Overall, the study demonstrated that the intermediate ponds of the Hambantota Salt Pan can serve as a useful source of halophilic microorganisms with potentially valuable extracellular enzymes. Further studies, including molecular identification such as 16S rRNA sequencing and quantitative enzyme characterization, are recommended to identify the isolates more accurately and evaluate their industrial potential.

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GROUP 3

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